Robust Control Via a Fuzzy PID Controller for Process Industry

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Abstract: In this paper a method for robust control via a fuzzy PID controller is designed with great disturbances and uncertainties taken into consideration. Conventional PID control method cannot guarantee ideal performance in most cases of great disturbance, time delay and nonlinear system, hence fuzzy control theory is introduced and fuzzy adaptive PID control is presented and analyzed in terms of stability and agility. Simulated experiments are given and analyzed using Matlab. The success of the proposed fuzzy adaptive PID control approach is demonstrated compared with traditional PID control.

Key words: Fuzzy Adaptive PID Controller; Robust Control; Process Industry

过程工业基于模糊自适应PID控制器的鲁棒控制

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摘 要: 本文在考虑大扰动和不确定性的情况下基于模糊PID控制器设计了一种鲁棒控制方法。由于传统PID控制方法对于大扰动、时延和非线性系统不能保证理想的性能，因而引入了模糊控制理论，提出了模糊自适应PID控制，分析了其稳定性和快速性；并在MATLAB环境下建立了模糊自适应PID控制系统仿真模型，通过比较模糊PID控制和传统PID控制的性能优劣，验证了模糊自适应PID控制的优越性。

关键词: 模糊自适应PID控制器；鲁棒控制；过程工业
中图分类号: TP273

1 Introduction

The proportional-integral-derivative (PID) control algorithm makes it the most widely used control strategy in the field of process industry because of its simplicity, robustness, and successful practical application. It is estimated that 90% of process industry control systems still use PID control as their main control strategy [1]. PID control behaves perfectly in the condition of stable process, little disturbance and linear system. However, in most process industry control systems, great disturbance, time delay and nonlinear factors are common cases. There exist many unstable processes in the chemical plants, even though most chemical processes are open-loop stable. The most common example is the batch chemical reactor, which has a strong nonlinearity due to the heat generation term in the energy balance [2]. Time-delayed unstable processes are commonly encountered in process industries and are difficult to control due to right-half plane (RHP) poles [3]. Although PID control has many advantages, it still exhibits dissatisfactory performance in many situations owing to its poor adaptivity. Designing a simple but efficient method for tuning the PID controller becomes an important issue.

In 1965, L. A. Zadeh combined classical collection

基金项目：国家自然科学基金（60805029）
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with multi-valued logic together, created fuzzy set theory and opened up a scientific solution to fuzzy issues. At present, fuzzy control has been applied in the field of chemical industry, machinery, metallurgy, food production so forth, which fully demonstrated its good control quality in complex, large scale and nonlinear systems [4]. Fuzzy adaptive PID control, combined with conventional PID control and fuzzy control, not only maintains the characteristics of simple structure, strong applicability and robustness of PID control, but also online adjust the parameters of PID controller with intelligent technology to adapt to the complicated control process. Present researches have proven good control quality for systems with little disturbance by fuzzy PID control. Some literature emphasizes the mathematical nature of fuzzy PID control [5]. Some focuses on the enhancement of dynamic performance in comparison with fuzzy PID control and conventional PID control [6, 7]. Others mainly give the design of fuzzy PID controller [8, 9]. However, more work needs to be investigated for robust control with fuzzy PID controller. So in this paper, a fuzzy adaptive PID controller is designed to deal with large disturbance, time-delay and nonlinear factors for robust control.

2 PID and Fuzzy Control

The mathematical model of PID control can be described as follows:

$$u(t) = K_p(e(t) + \frac{1}{T_i}\int_0^t e(\tau)d\tau + T_d\frac{de(t)}{dt})$$

$e(t)$ is the error input of PID controller. $u(t)$ is the output. K_p , K_i and K_d are the coefficients of proportional, integral and derivative parts, respectively. The tuning of K_p, K_i and K_d is significantly critical for the control quality of PID control. Although calculation based on transfer function can figure out the precise values of these coefficients, the transfer function itself is always imprecise or uncertain in most cases. So it is almost impossible for us to get accurate results of the three parameters in real industry process. Therefore, some engineering tuning method based on long period of experience is put forward. Among them, Z-N method and critical proportional band method are most frequently used approaches to gain the values of three parameters. However, the tuning processes and final results depend on operators' experience to great extent. It is difficult to find a group of optimal controller parameters and to realize the online adaptive tuning for PID parameters [10]. Although PID control has unparalleled advantages, it still exists great deficiencies under extreme conditions. Therefore, new measures are to be introduced in to improve the robustness of PID controller.

Since 1960s, the rise and development of artificial intelligent gained wide attention in the field of control theory. Especially, fuzzy control proposed by Lofty Zadeh in 1965 emerged as a tool to deal with uncertain or qualitative decision-making problems. In fuzzy logic, mathematical equations are replaced by linguistic descriptions to define the relationship between the input information and the output action [11]. Fuzzy logic attempts to model computer reasoning on this kind of imprecision and vagueness like human reasoning. Through fuzzy logic, a system not only can represent such imprecise concepts as *Fast, Tall,* and *Expensive*, but through a set of sound mathematical principles, it can also use these concepts to make deductions about a system. Fuzzy logic is good at dealing with imprecise or vague inputs and it has the following main merits: (1) Fuzzy logic has been proven to be very effective in modeling complex, often nonlinear, systems. This modeling is usually very rapid, proceeds without a mathematical model, and is based on an approximate representation of the control surface and its behavior as the system is fed various input values [12]. (2) In engineering systems, fuzzy logic provides a convenient and user-friendly front-end to develop control programs, helping designers to concentrate on the functional objectives instead of mathematics.

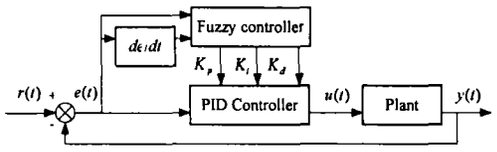

Figure 1 Principle of Fuzzy adaptive PID controller

3 Fuzzy adaptive PID control

Fuzzy adaptive PID control is constructed on the basis of both fuzzy control theory and conventional PID control

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advantages. Fuzzy control needs not be built based on precise mathematical model, while PID keeps its robustness and applicability. Fuzzy PID control not only improves steady-state accuracy but also increases the adaptability for complex process industry and enhances the performance of system control.

3.1 Structure of fuzzy adaptive PID

The structure of fuzzy PID controller is shown in Figure 1. Fuzzy adaptive PID controller consists of two parts: Fuzzy controller and PID controller. The inputs of fuzzy controller include the error and the change of error. It inferences and processes input data according to certain fuzzy rules. The outputs of fuzzy controller are crisp values which act on PID controller. Its core idea is to first find out the fuzzy relationship between the three parameters and the error *e*, the change of error *ec*. By continuously detecting *e* and *ec*, the 3 parameters are changed online according to fuzzy rules, so that they can meet the control demands to improve the characteristics of control systems. The parameter tuning formula of fuzzy PID controller is shown as follows:

$$k_p = k_p^{'} + \Delta k_p \quad k_i = k_i^{'} + \Delta k_i \quad k_d = k_d^{'} + \Delta k_d$$

$k_p^{'}, k_i^{'}, k_d^{'}$ are the initial values of PID controller parameters. Δk_p, Δk_i and Δk_d are the outputs of fuzzy controllers. k_p, k_i, k_d are final outputs of fuzzy PID controller. The dynamic tuning of k_p, k_i, k_d makes it possible for PID controller to adapt to the environment simultaneously and adjust its parameters online automatically.

3.2 Fuzzy rules

The principle of tuning parameters usually derives from expert experience. At the same time, we need to adjust the rules repeatedly until they match the fuzzy adaptive PID control strategy perfectly. According to expert experience and repeated tuning, we can get the following table of fuzzy rules.

Fuzzy PID control not only keeps the advantage of conventional PID control, but also integrates the trait of fuzzy control that it does not need to rely on accurate mathematical model. It can online adjust parameters of PID controller. Its major merits are twofold: (1) Good Static and dynamic performance. The introduction of fuzzy PID control greatly improves the stability and speed and system performance is optimized. (2) Robustness. With the combination of fuzzy logic, system robustness is significantly enhanced. The resistance to external disturbance is strengthened and the impact of structural change is weakened.

Table 1 Fuzzy rules of k_p

e \ ec (Δk_p)	NB	NM	NS	ZO	PS	PM	PB
NB	PB	PB	PM	PM	PS	ZO	ZO
NM	PB	PB	PM	PS	PS	ZO	NS
NS	PM	PM	PM	PS	ZO	NS	NS
ZO	PM	PM	PS	ZO	NS	NM	NM
PS	PS	PS	ZO	NS	NS	NM	NM
PM	PS	ZO	NS	NM	NM	NM	NB
PB	ZO	ZO	NM	NM	NM	NB	NB

Table 2 Fuzzy rules of k_i

e \ ec (Δk_i)	NB	NM	NS	ZO	PS	PM	PB
NB	NB	NB	NM	NM	NS	ZO	ZO
NM	NB	NB	NM	NS	NS	ZO	ZO
NS	NB	NM	NS	NS	ZO	PS	PS
ZO	NM	NM	NS	ZO	PS	PM	PM
PS	NM	NS	ZO	PS	PS	PM	PB
PM	ZO	ZO	PS	PS	PM	PB	PB
PB	ZO	ZO	PS	PM	PM	PB	PB

Table 3 Fuzzy rules of k_d

e \ ec (Δk_d)	NB	NM	NS	ZO	PS	PM	PB
NB	PS	NS	NB	NB	NB	NM	PS
NM	PS	NS	NB	NM	NM	NS	ZO
NS	ZO	NS	NM	NM	NS	NS	ZO
ZO	ZO	NS	NS	NS	NS	NS	ZO
PS	ZO	ZO	ZO	ZO	ZO	ZO	ZO
PM	PB	NS	PS	PS	PS	PS	PB
PB	PB	PM	PM	PM	PS	PS	PB

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4 Simulated Experiments

Now a typical second-order water level control system is adopted as an example to verify the effectiveness of PID control and fuzzy adaptive PID control with the software of MATLAB/Simulink. The distinction of the results of two control methods verifies the robust advantages of fuzzy PID control.

4.1 PID control model

A typical boiler is set as the controlled plant. The transfer function of the input water flux and water level can be expressed as: $G(s)=\dfrac{0.05}{8.5s^2+s}$. The closed loop control system based on conventional PID control is constructed first. It is shown in Figure 2. According to critical proportion band method, the PID controller initial values are set as: $k_p^{'}=300$, $k_i^{'}=0.3$, $k_d^{'}=280$.

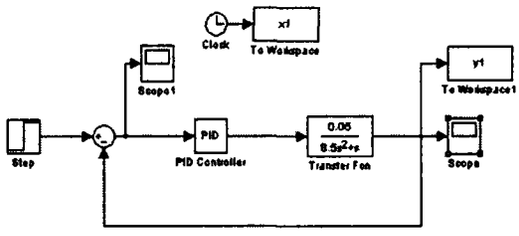

Figure 2 Conventional PID control model

4.2 Fuzzy Adaptive PID model

Fuzzy adaptive PID control system comprises of the PID subsystem and fuzzy controller. The PID subsystem is built as figure 3. The fuzzy controller is set as follows. "And method" is set as "min". "Or method" is set as "max". "Implication" is set as "max". "Aggregation" is set as "max". At last, "Defuzzification" is set as "centroid". The input parameters are "*e*" and "*ec*", which represent error and change of error. The output parameters are $\Delta k_p, \Delta k_i, \Delta k_d$. The domains of the five parameters are all set as {-6,-5,-4,-3,-2,-1,0,1,2,3,4,5,6}. Each parameter owns the following fuzzy sets {NB, NM, NS, ZO, PS, PM, PB}.

The triangle membership function is simply structured and has high computation efficiency which is fit for real time control and adaptive control. Accordingly, the input and output parameters in this paper choose the triangle curve as their membership function shown in Figure 4.

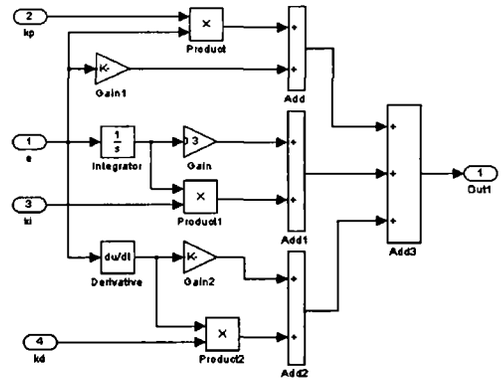

Figure 3 PID subsystem

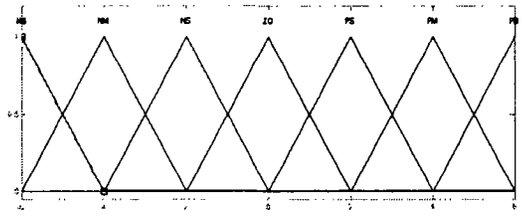

Figure 4 Triangle membership function

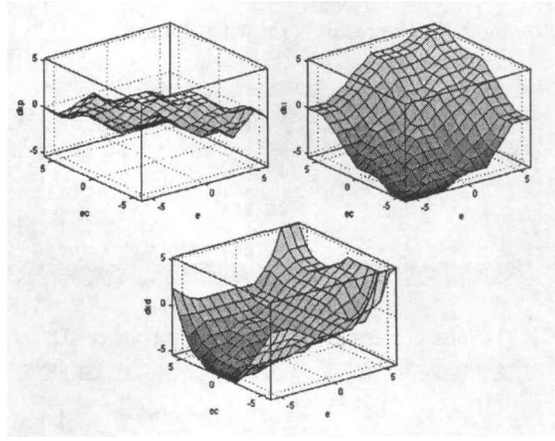

Figure 5 Surface viewer

The fuzzy rules are set according to the previous rules of table 2, 3 and 4. Surface Viewer shows the results of the rules as Figure 5. By observing the optimal response state during the tuning process, the appropriate quantitative factors are $k_e=0.005$, $k_{ec}=0.005$, and scale factors are $k1=0.001$, $k2=0.05$, $k3=0.001$. Finally, the complete fuzzy PID control system is constructed in figure 6.

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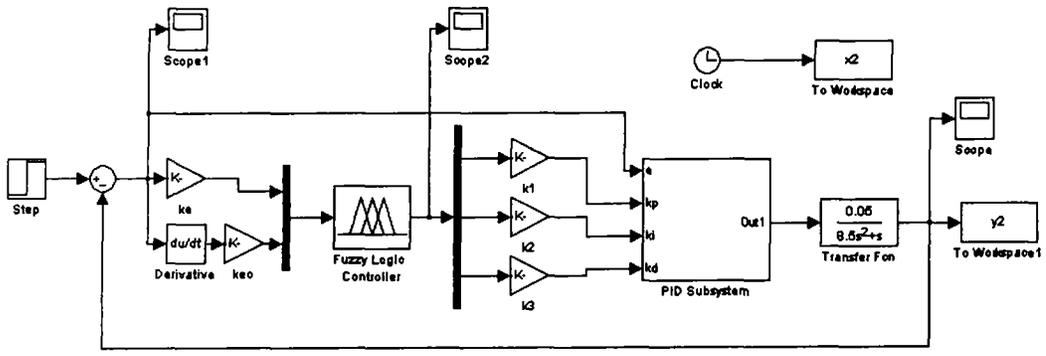


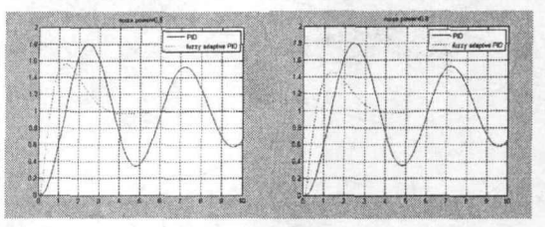
Figure 6 Fuzzy adaptive PID control system

4.3 Results Analysis

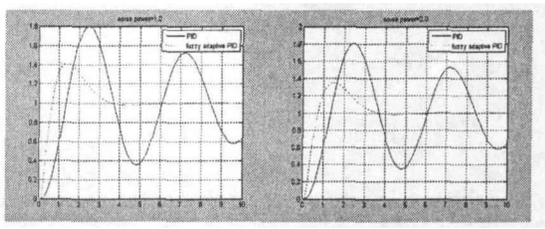
After establishing the PID control system and fuzzy PID control system, the performances of the two control strategy are compared under the circumstance of (1) system with noise disturbance; (2) system with time delay; (3) nonlinear system. The sample time is set as 0.01s.

4.3.1 System with noise disturbance

White noise disturbance is introduced into PID control and fuzzy PID control systems. The initial noise power is set as 0.5; increase the power of noise gradually to 0.8, 1.0 and 2.0. Compare the advantages and disadvantages of the two control method. The result is shown as Figure 7.



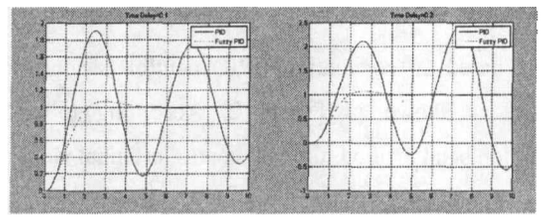
(1)Noise power=0.5 (2)Noise power=0.8



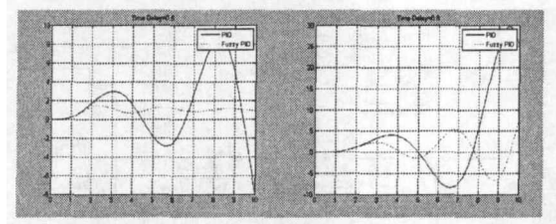
(3)Noise power=1.0 (4)Noise power=2.0

Figure 7 System with noise disturbance

Through Figure 7, it is clear that fuzzy PID control is able to step into stable state in a very short time even with the noise power twice stronger than the input power. On the contrary, the traditional PID can hardly become stable in short time. Its overshoot is also much greater than fuzzy PID control.



(1)Time delay=0.1 (2)Time delay=0.2



(3)Time delay=0.5 (4)Time delay=0.8

Figure 8 System with time delay

4.3.2 System with time delay

Time delay is taken into consideration in PID control and fuzzy PID control. At first, time delay is set as 0.1s. Gradually, it increases to 0.2s, 0.5s and 0.8s. The results are shown as in Figure 8. Apparently, when time delay is less than 0.5s, fuzzy PID control can lead to a stable end. However, traditional PID control diverges quickly even with a time delay of 0.2s. The figures demonstrate that fuzzy adaptive PID control owns better control ability in systems with time delay.

4.3.3 System with nonlinear saturation

Nonlinear saturation is introduced in control system now. Upper and lower limit is set as 20, 10, 5 and 3. The results are shown as Figure 9. We can obviously find out

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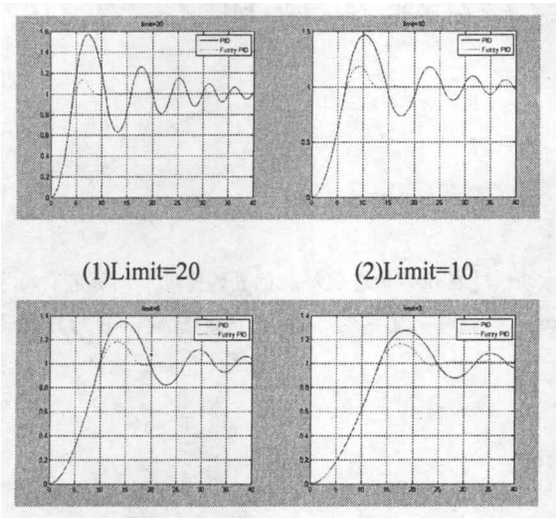
that fuzzy PID control also behaves better in system with nonlinear saturation. On the contrary, great overshoot reflects traditional PID control's weakness.



(1)Limit=20 (2)Limit=10

(3)Limit=5 (4)Limit=3

Figure 9 System with nonlinear saturation

5 Conclusions

In this paper a robust control method based on fuzzy adaptive PID controller is designed and realized by combining fuzzy control with traditional PID control. This method not only retains the PID controller's advantages of simple structure, applicability and robustness, but also agilely tunes the parameters of PID controller online for complex process control. A typical second-order control system is taken as an example to simulate the effectiveness of fuzzy PID control under the circumstances of disturbance, time delay and nonlinear saturation. By comparing the performances of traditional PID controller and fuzzy adaptive PID controller, the success of the presented method is verified.

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